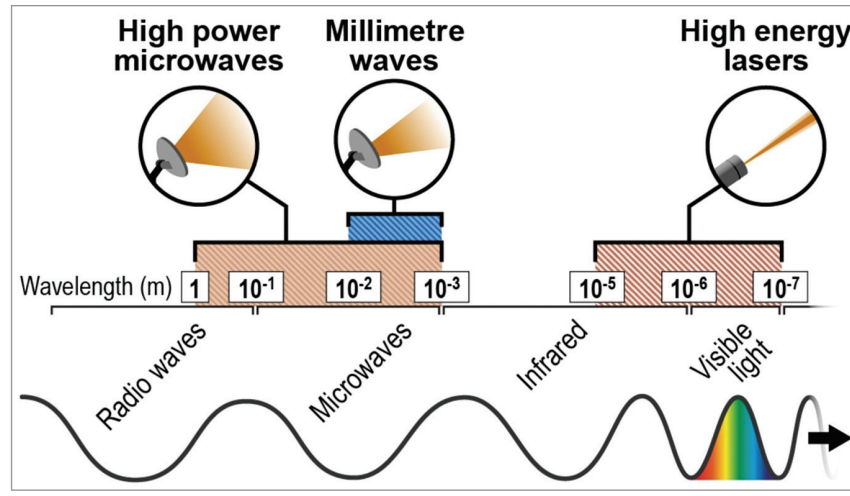
They offer surgical precision and a nearly unlimited number of shots. However, their performance is weakened by environmental factors like fog, rain, or dust that scatter the beam.

High-power microwaves (HPMs). HPMs work by firing intense pulses of radio-frequency energy to 'fry' the target's electronics. The energy surge enters through openings like antennae or penetrates 'back-door' paths such as unshielded wiring and tiny seams in the casing. Once inside, the surge overloads microchips and sensors, crashing the flight logic or permanently destroying the hardware. Since the pulses cover a wide area, they can disable multiple threats using a single shot, creating an invisible 'no-fly zone'.

Particle beam weapons. Particle beam weapons are the most advanced in terms of directed-energy technology. They work by accelerating streams of atomic or subatomic particles, such as protons, ions, or electrons, to near-light speeds and directing them to a target. On impact, the particles deliver immense energy that superheats materials, ionises matter, and disrupts electronic systems simultaneously. Particle beam weapons have the capability to intercept missiles during launch, disable satellites in orbit, or neutralise targets with precision. However, they need enormous power, complex particle accelerators, and advanced focus systems to keep the particle beams coherent over a long distance, making them far less practical than lasers or microwave-based weapons. They continue to remain at an experimental level.



HPM weapons have the broadest beam pattern and HELs have the narrowest beams (Source: US Government Accountability Office)

Acoustic weapons. Acoustic weapons use specialised speakers, called transducers, to beam amplified sound waves at a target to create a 'sound barrier'. These sound waves operate at frequencies or volumes unbearable to a human ear, causing discomfort, disorientation, or pain. Such weapons are used as a non-lethal option to manage crowds or keep unauthorised naval vessels away.

Hybrid systems. Hybrid systems combine technologies to create a 'layered' defence mechanism. For example, a system may use an HPM pulse to disable the drone's navigation system and then apply a laser beam to destroy it physically. Another variant may integrate acoustic waves to disperse or deter the enemy and then deploy laser systems to disable their equipment.

DEWs: Advantages

DEWs are a transformative technology offering faster, precise, and cost-effective solutions to counter evolving threats and providing tactical and psychological advantages. A single system can be scaled from a handheld tactical device to a large naval platform. Travelling at the speed of light, they eliminate the 'time-of-flight' associated with conventional projectiles, reducing reaction time against high-speed targets like hypersonic missiles and drones. They allow operators to neutralise targets in sensitive areas, such as urban power plants or airports, causing minimal collateral damage. They offer an 'infinite magazine', because they draw power from onboard generators or energy modules rather than physical munitions, eliminating the logistical

Box 1: The principle of electromagnetics

Electromagnetics is the interaction between electric and magnetic fields to propagate energy. Moving charges create magnetic fields, while fluctuating magnetic fields induce electric fields, enabling energy to travel as electromagnetic waves. While these waves are commonly used for radio communication or heating in microwave ovens, DEWs scale these waves for defence applications.

TABLE 1
OPERATIONAL ADVANTAGES

Type of DEW	Advantages	Challenges
High-energy lasers	Precision, speed-of-light strike, and reusable shots	High power demand, cooling, and weather effects
High-power microwaves	Effective against electronics and swarm defence	Limited range and shielding countermeasures
Particle beam weapons	Potential deep penetration and futuristic capability	Immense power requirement, stability, and portability
Acoustic/hybrid	Non-lethal options, and niche applications	Limited destructive power and short effective range